

Performance and Classifier Validation Testing

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Performance and Classifier Validation Testing
Project Phase	Phase 6: Project Testing
Submitted By	Team Leader: Palli Dharani Team Members: Yarabarla Manidweep Saikiran Kudipudi Sita Rama Raju Datla Thota Leela Sai Krishna
Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

1. Model Selection Benchmarks

Model Classifier	Test Accuracy Score	Cross-Validation Mean (cv=5)	Variance Ratio
Decision Tree Classifier	78.4%	77.1%	+/- 2.1%
Random Forest Classifier	81.2%	80.5%	+/- 1.8%
K-Nearest Neighbors (KNN)	79.5%	78.9%	+/- 1.5%
XGBoost Classifier	85.6%	84.8%	+/- 1.1% (Selected Model)

2. Inference Pipeline Classification Performance Report

```
[ ]  #importing and building the Xg boost model
def XGB(X_train,X_test,y_train,y_test):
    model = GradientBoostingClassifier()
    model.fit(X_train,y_train)
    y_tr = model.predict(X_train)
    print(accuracy_score(y_tr,y_train))
    yPred = model.predict(X_test)
    print(accuracy_score(yPred,y_test))

[ ] #printing the train accuracy and test accuracy respectively
XGB(X_train,X_test,y_train,y_test)
```

Figure: XGBoost Classification Matrix Model Performance Logs

3. Server Load Performance Testing

Metric	Target Limit Constraint	Observed Value Passed
Page Load Response Time	Max 1500 ms	Average 180 ms
Prediction Inference CPU Time	Max 500 ms	Average 85 ms
Concurrent Session Handling	50 requests/sec limit	Passed without memory overhead increases